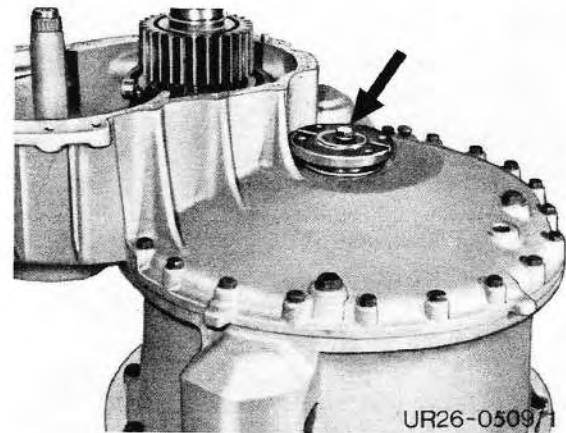
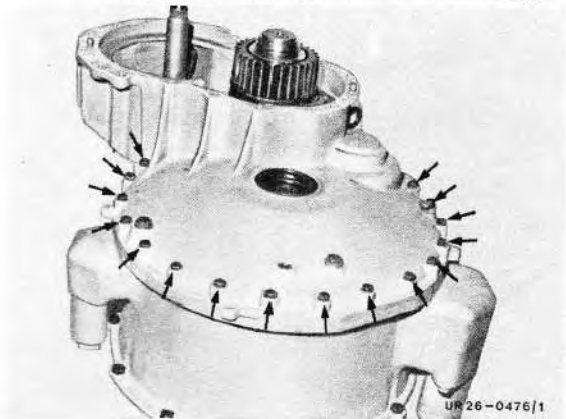


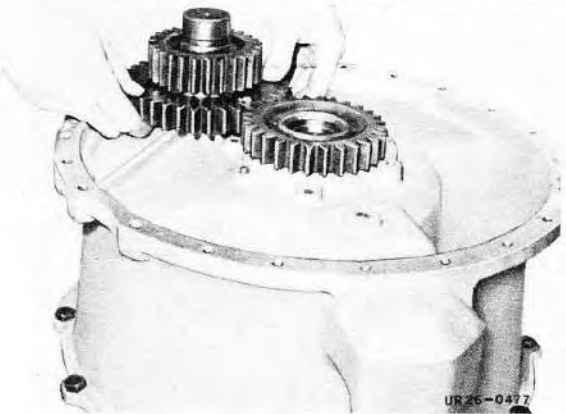
- 12 Remove retaining plate, remove output flange and engine-mounted pto input.



- 13 Remove transmission casing, remove, if necessary, shaft seal and roller bearing.

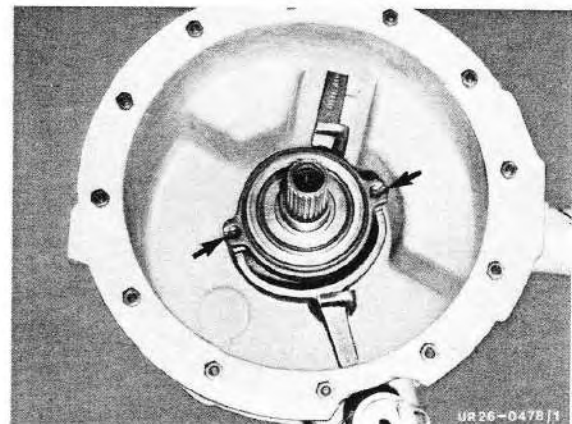


- 14 Remove idler gears.
15 Turn clutch housing through 180°.

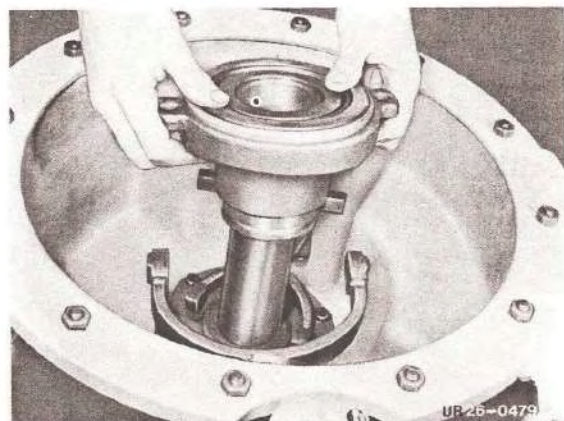


Applicable for engine-mounted pto

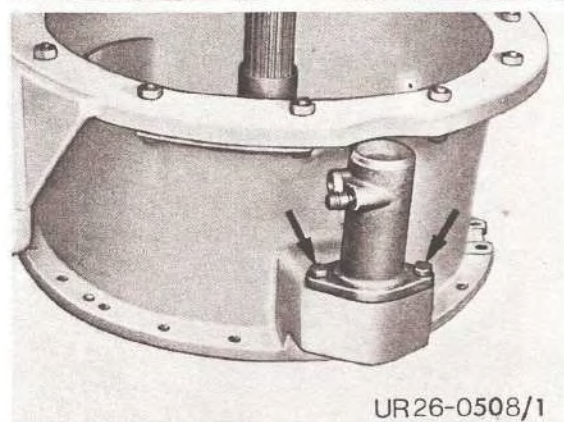
- 16 Remove pre-formed springs on the pto release lever.



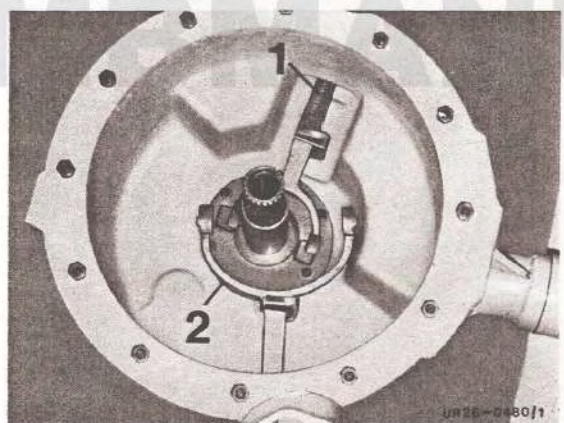
SA 35 737



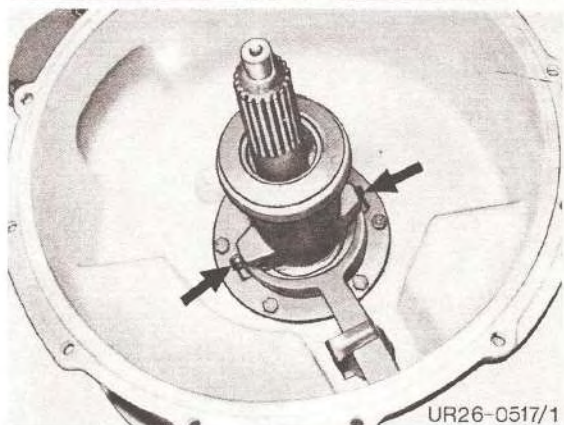
- 17 Remove pre-formed springs on the drive clutch release lever and remove release bearing.



- 18 Remove clutch slave cylinder for drive and pto clutch.



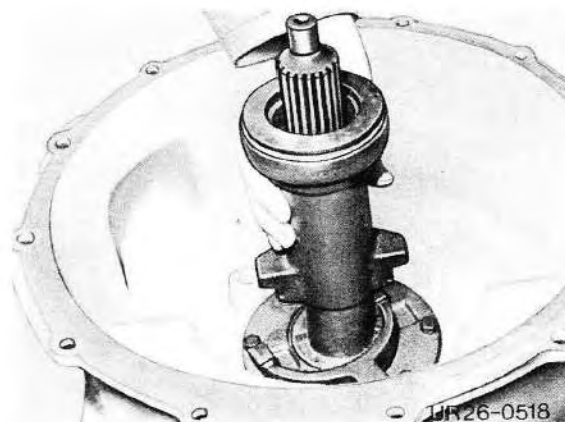
- 19 Remove release lever for pto clutch (2) and drive clutch (1).



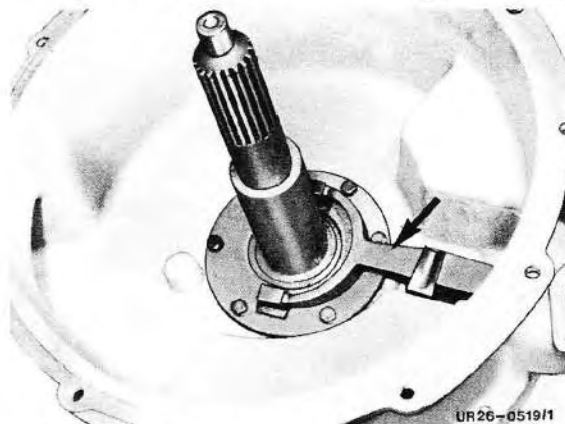
Applicable for transmission-mounted pto

- 20 Remove pre-formed spring on pto release lever.

- 21 Remove release bearing.

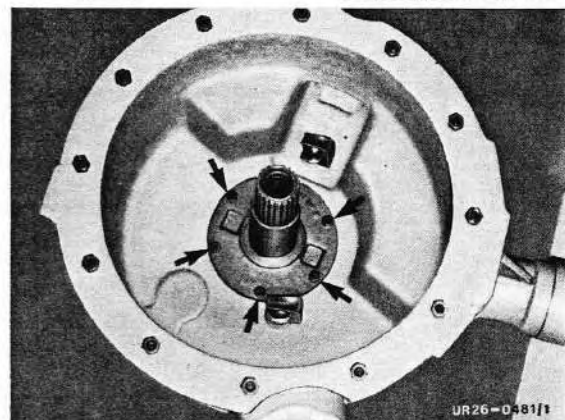


- 22 Remove release lever for drive clutch.

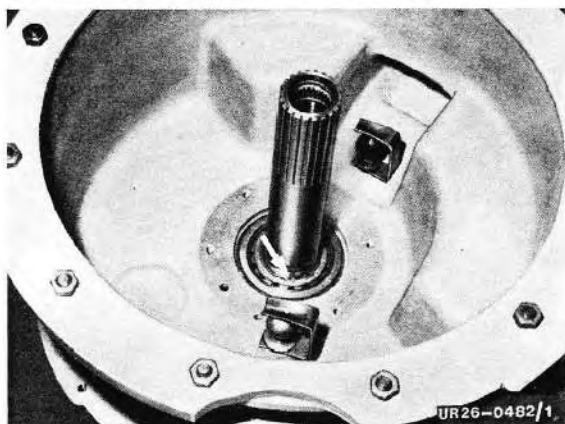


Applicable for engine- and transmission-mounted pto's

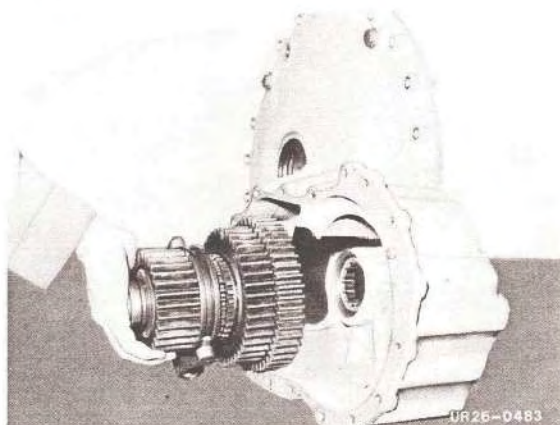
- 23 Unscrew fastening bolts, take off guide flange and remove shaft seal.



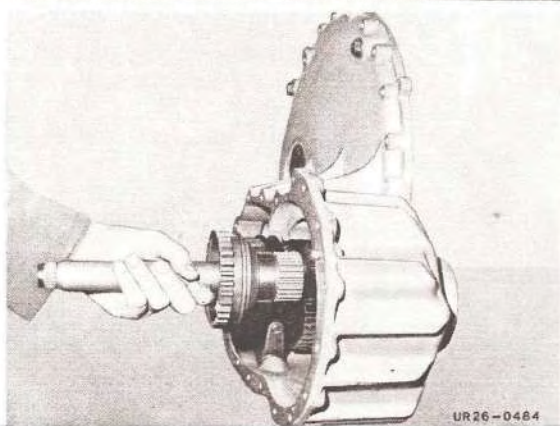
- 24 Take off circlip and remove engine-mounted pto input gear.



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25 Remove input shaft with selector fork.
Disassemble input shaft, see 3.1/1.

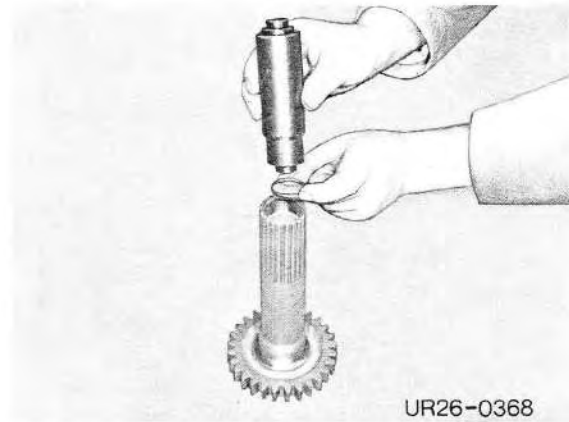


26 Remove output shaft.
27 Clean all parts, check and replace if necessary.
28 Fit pto transmission,
see 2.1/3.

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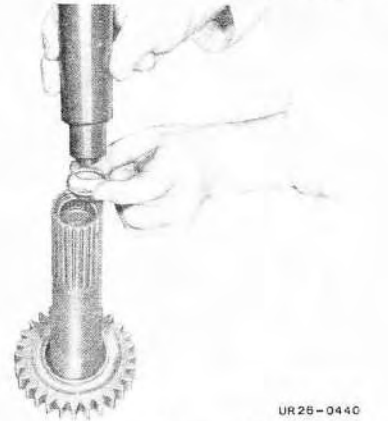
Assembly

- 1 Press in roller bearing into clutch housing.
- 2 Press in spacer ring with pressing tool No. 6.



UR26-0368

- 3 Fit needle bearing with grease No. 5.
- 4 Press in spacer ring with pressing tool No. 6.



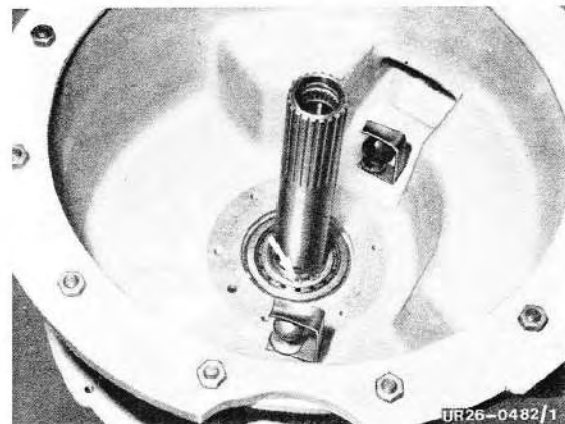
UR 26-0440

- 5 Coat shaft seal with grease No. 2 and press in with pressing tool No. 6.



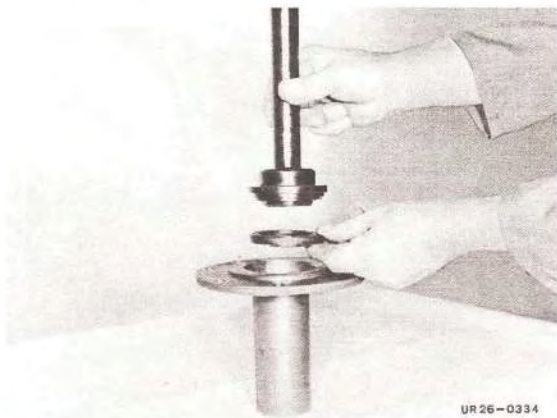
UR26-0369

- 6 Fit engine-mounted pto input gear and secure.

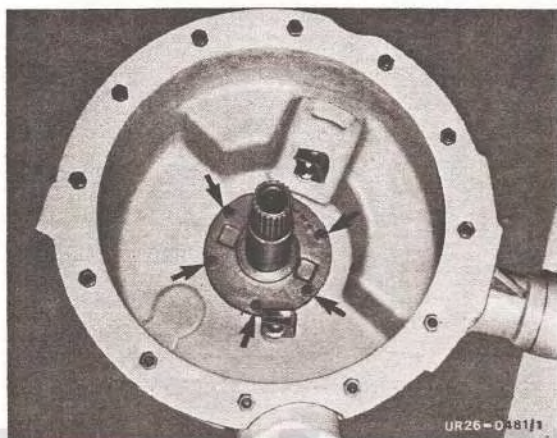


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SA 35 737

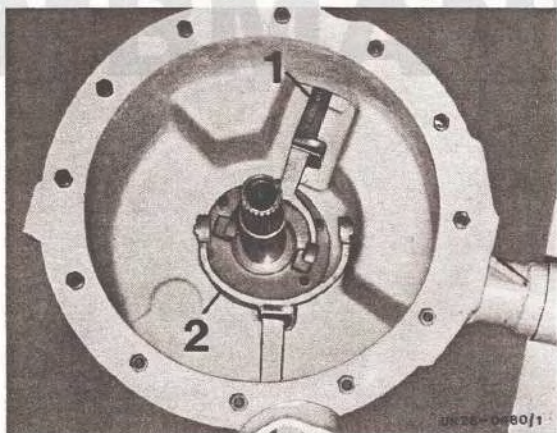


- 7 Coat shaft seal with grease No. 2 and press in with pressing tool No. 8.



- 8 Coat guide flange with sealant No. 1 and fit.

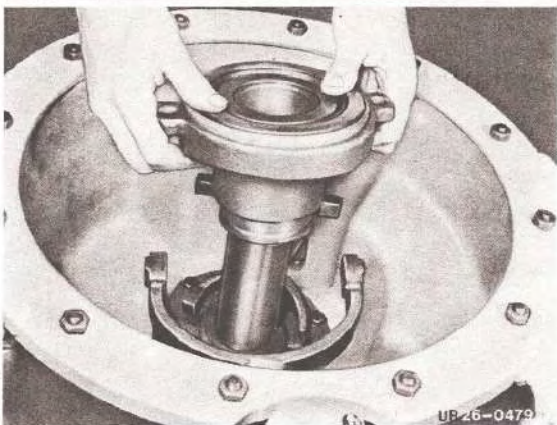
Note: Watch out for oilway.



Applicable for engine-mounted pto

- 9 Fit release lever for drive clutch (1) and pto clutch (2).

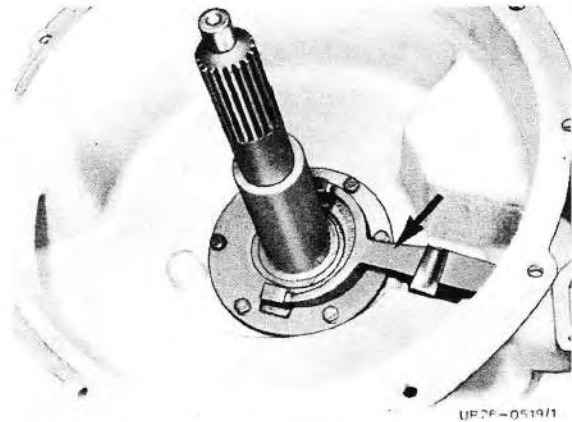
- 10 Press deep groove ball bearing into release bearing.



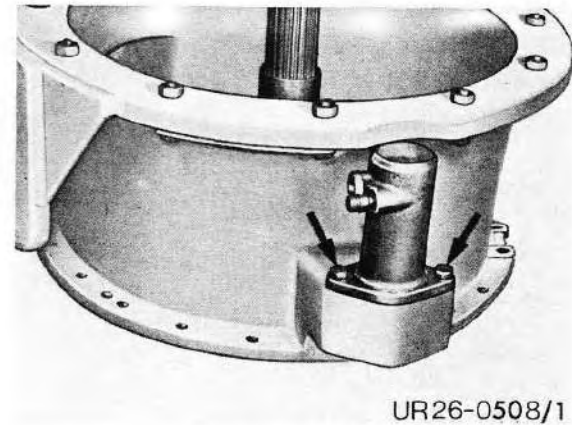
- 11 Fit release bearing and fit pre-formed springs to the drive clutch release lever and the pto release lever.

Applicable for transmission-mounted pto

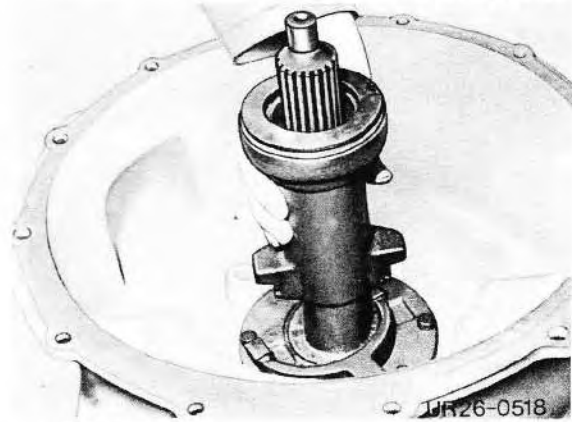
12 Fit release lever for drive clutch.



13 Fit clutch slave cylinder.

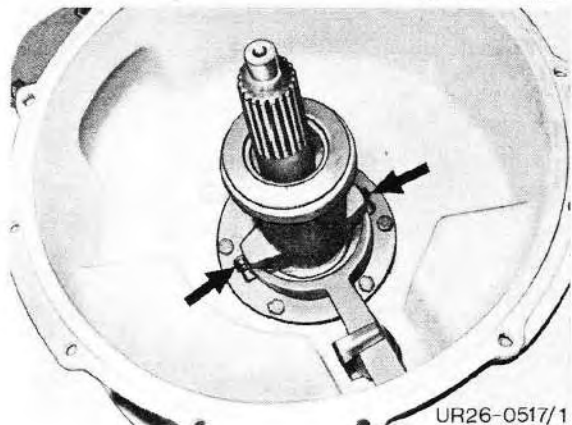


14 Fit release bearing.

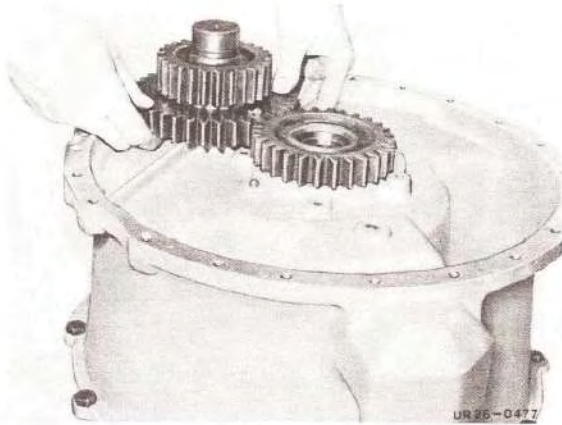


15 Fit pre-formed springs in pto release lever.

16 Turn clutch housing through 180°.

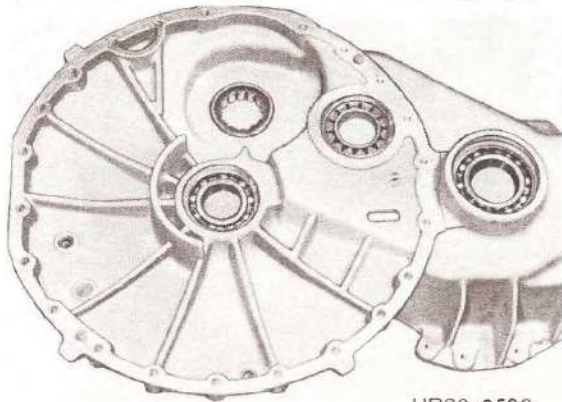


SA 35 737

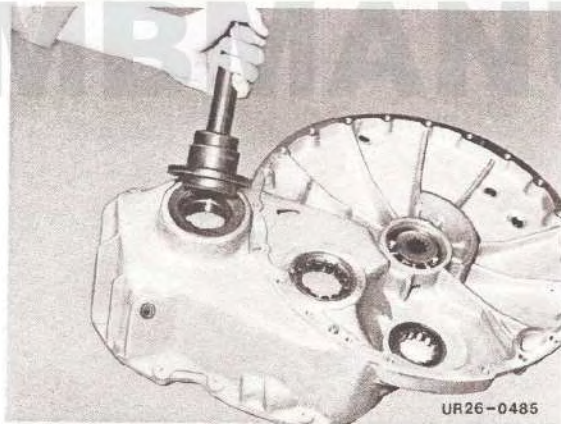


Applicable for engine- and transmission-mounted pto's

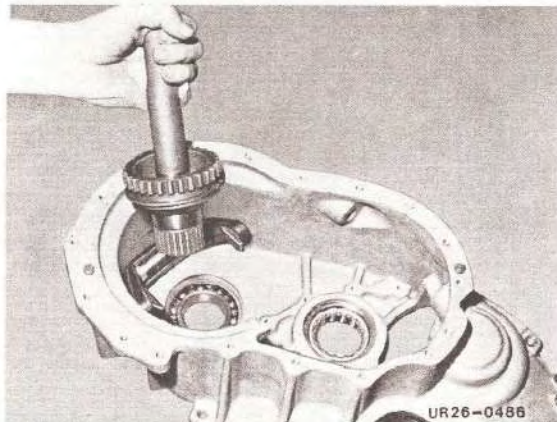
17 Press roller bearing on to idler gear and fit idler gear.



18 Fit roller bearing in transmission housing and secure with circlip. Press in dog ring.



19 Coat shaft seal with grease No. 2 and press in with pressing tool No. 3.

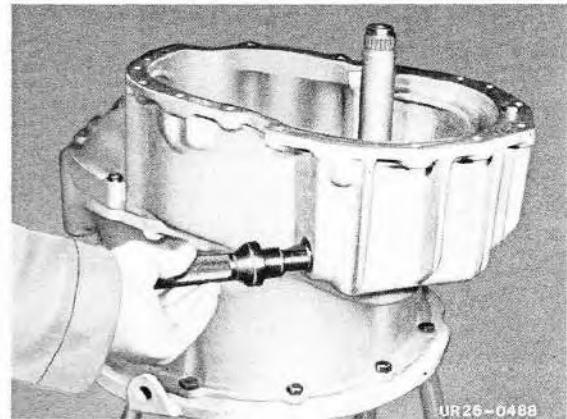


20 Fit output shaft.

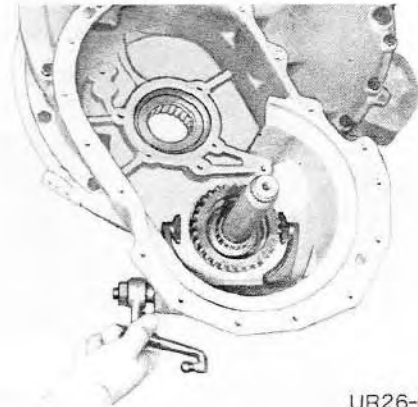
- 21 Fit transmission casing with gasket.



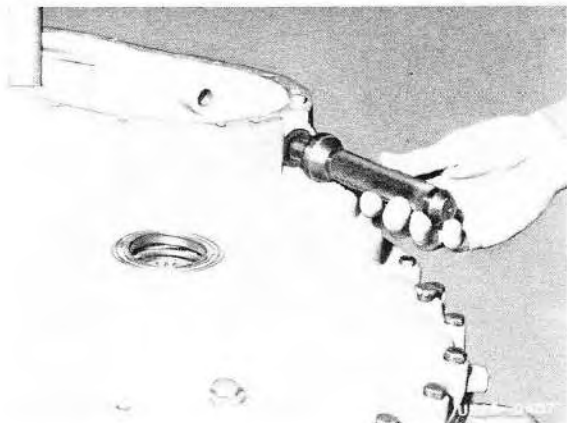
- 22 Coat shaft seal with grease No. 2 and press in with pressing tool No. 7.



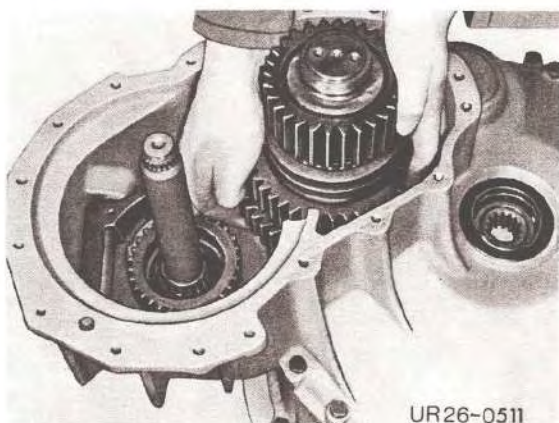
- 23 Put selector fork in position of front pto input and fit selector shaft according to markings.



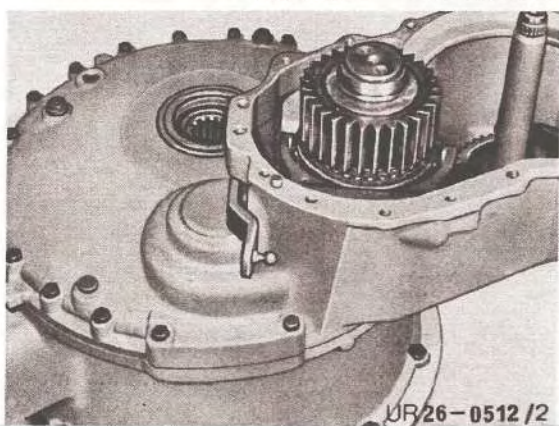
- 24 Coat shaft seal with grease No. 2 and press in with pressing tool No. 7.



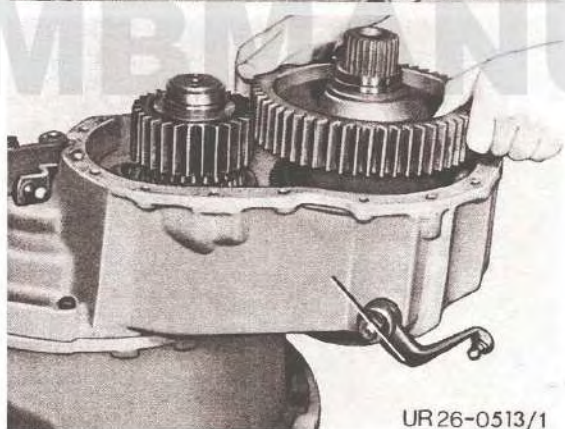
SA 35 737



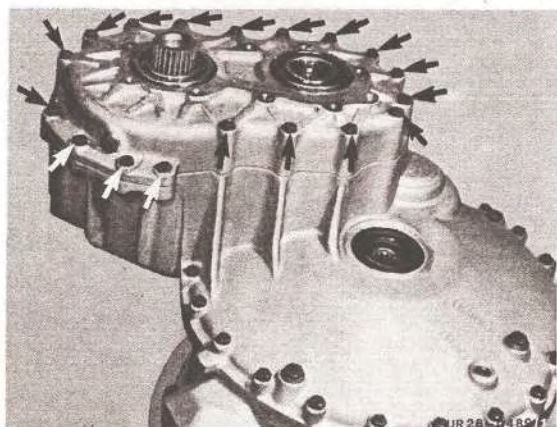
25 Fit input shaft with selector fork. Rebuild input shaft, see 4.1/1.



26 Fit selector shaft horizontally in neutral (follow markings) and secure with dowel pin.

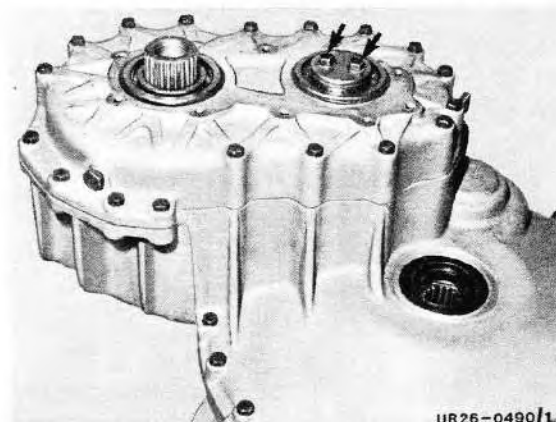


27 Put sliding sleeve in position of front pto input and fit output gears. Rebuild output gears, see 5.1/2.

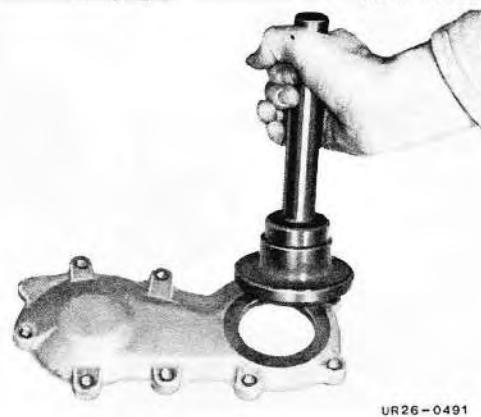


28 Fit casing cover with gasket.

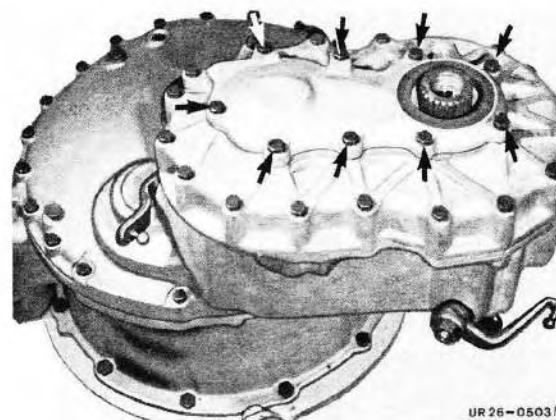
- 29 Press roller bearing on to input shaft.
- 30 Coat retaining plate with sealant No. 1 and screw tight.
- 31 Press roller bearing on to output gears.



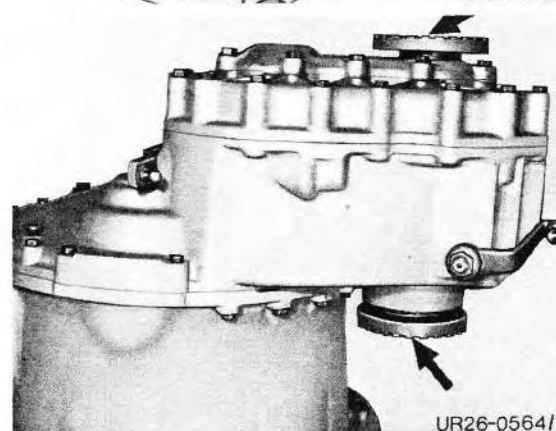
- 32 Coat shaft seal with grease No. 2 and press in with pressing tool No. 3.



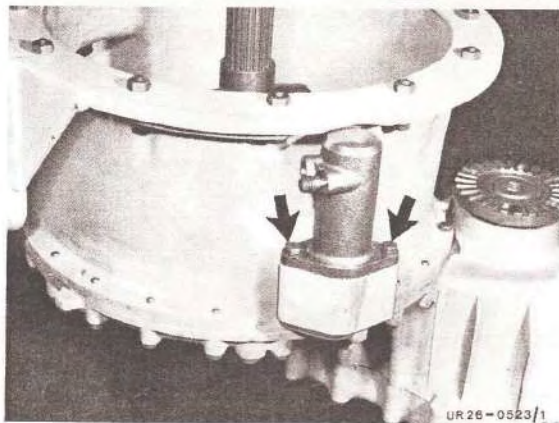
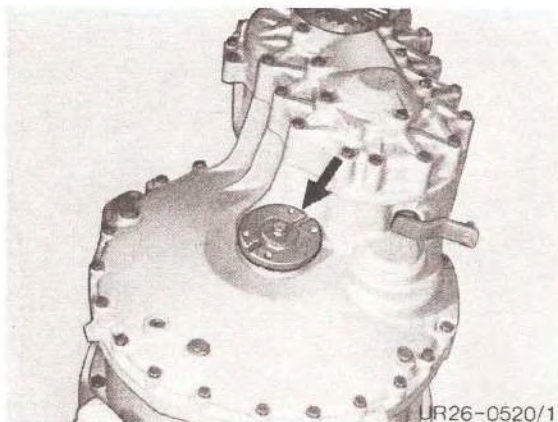
- 33 Fit nilos ring and screw on cover plate with gasket.



- 34 Coat both sets of teeth on the output shaft with running-in paste No. 3, fit flange, coat sealing bolt with locking material No. 4, screw tight and secure with centre punch.



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35 Coat tothing on engine-mounted pto with running-in paste No. 3. Fit output flange and screw tight with retaining plate.

36 Turn pto transmission through 180°.

Note: Use self-locking bolts only once.

37 Fit drive clutch slave cylinder.

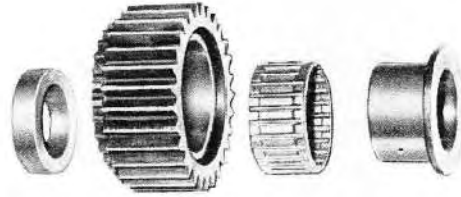
Note: Adjusting and fitting the pto slave cylinder is only possible when the pto transmission is fitted. Adjust pto slave cylinder, see 29.10-5,1/1.

38 Fit pto transmission, see 2.1/3.

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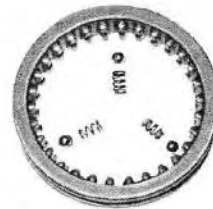
Disassembly

- 1 Pull off inner race of roller bearing and spacer.
- 2 Remove gear for 540/min, bearing bush, needle bearing and distance ring.



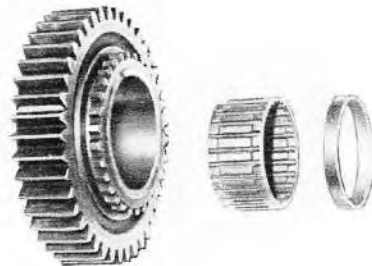
UR 26-0493

- 3 Take off sliding sleeve.
- Note:** Springs and balls will spring out of the guide.
- 4 Pull off dog ring.



UR 26-0494

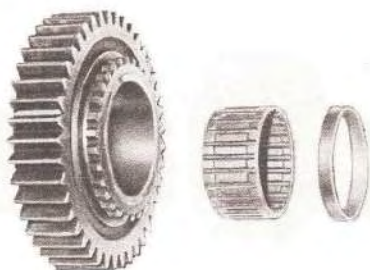
- 5 Remove distance ring, gear for 1 000/min and needle bearing.
- 6 Clean all parts, check and change if necessary.



UR 26-0495

Assembly

- 1 Fit needle bearing, gear for 1 000/min and distance sleeve on input shaft.



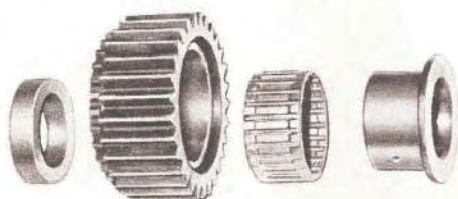
UR26-0495

- 2 Press on dog ring.
- 3 Fit sliding sleeve with springs and balls.



UR26-0494

- 4 Fit distance ring, gear for 540/min, needle bearing and bearing bush.

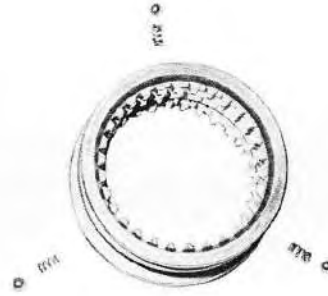


UR26-0493

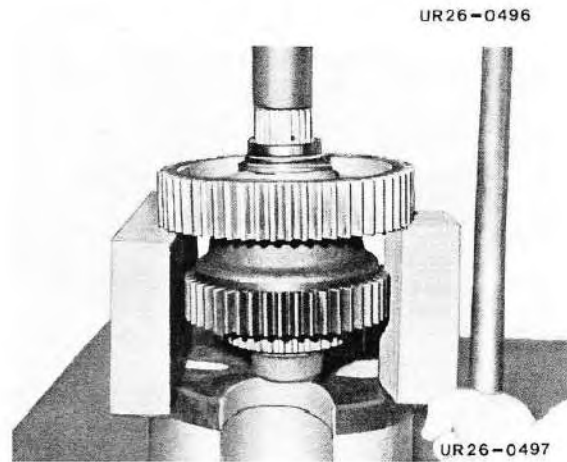
Disassembly

- 1 Remove sliding sleeve.

Note: Springs and balls will spring out of the guide.



- 2 Press sleeve shaft from output gears.

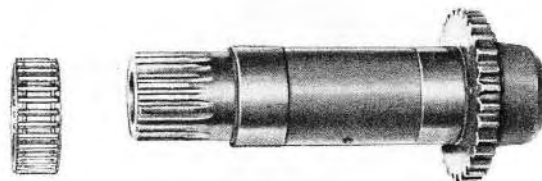


- 3 Remove inner race of roller bearing, distance ring and needle bearing.



UR26-0498

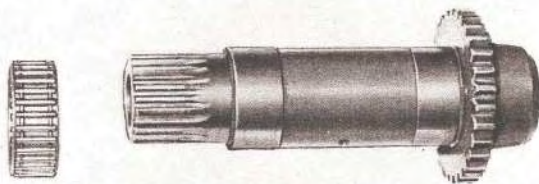
- 4 Remove needle bearing from sleeve shaft.



UR26-0499

Assembly

- 1 Push needle bearing on to sleeve shaft.



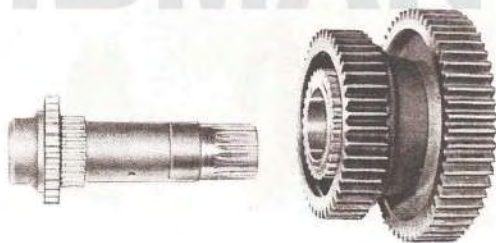
UR26-0499

- 2 Fit needle bearing, distance ring and inner race of roller bearing.



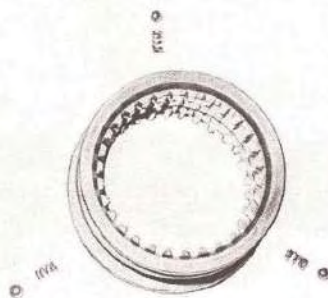
UR26-0498

- 3 Fit sleeve shaft into output gears.



UR26-0500

- 4 Fit sliding sleeve with springs and balls.



UR26-0496

Contents

Chapter	Page
1 General	
Installation survey	1.1/1
General view	1.1/1
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Sectional view, engine and PTO transmission shaft	1.2/2
Capacities	1.3/1
Special tools	1.3/1
Necessary materials	1.3/2
Tightening torques	1.4/1
Sectional view	1.5/1
2 Removal and installation of PTO transmission	2.1/1
3 Disassembly and assembly of PTO transmission	3.1/1
4 Disassembly and assembly of input shaft	4.1/1

Installation survey

Chassis Model	Sales designation	PTO transmission 540/min ¹⁾		Dir. of rotation	Installation
		Version	PTO		
435.111	U 1700 L	Transmission pto shaft	at front or at front and rear	clockwise	SA 35 737
.113	U 1700 L/38				
.115	U 1300 L				
.117	U 1300 L/37				

1) with fording ability

General view



UR26-0921

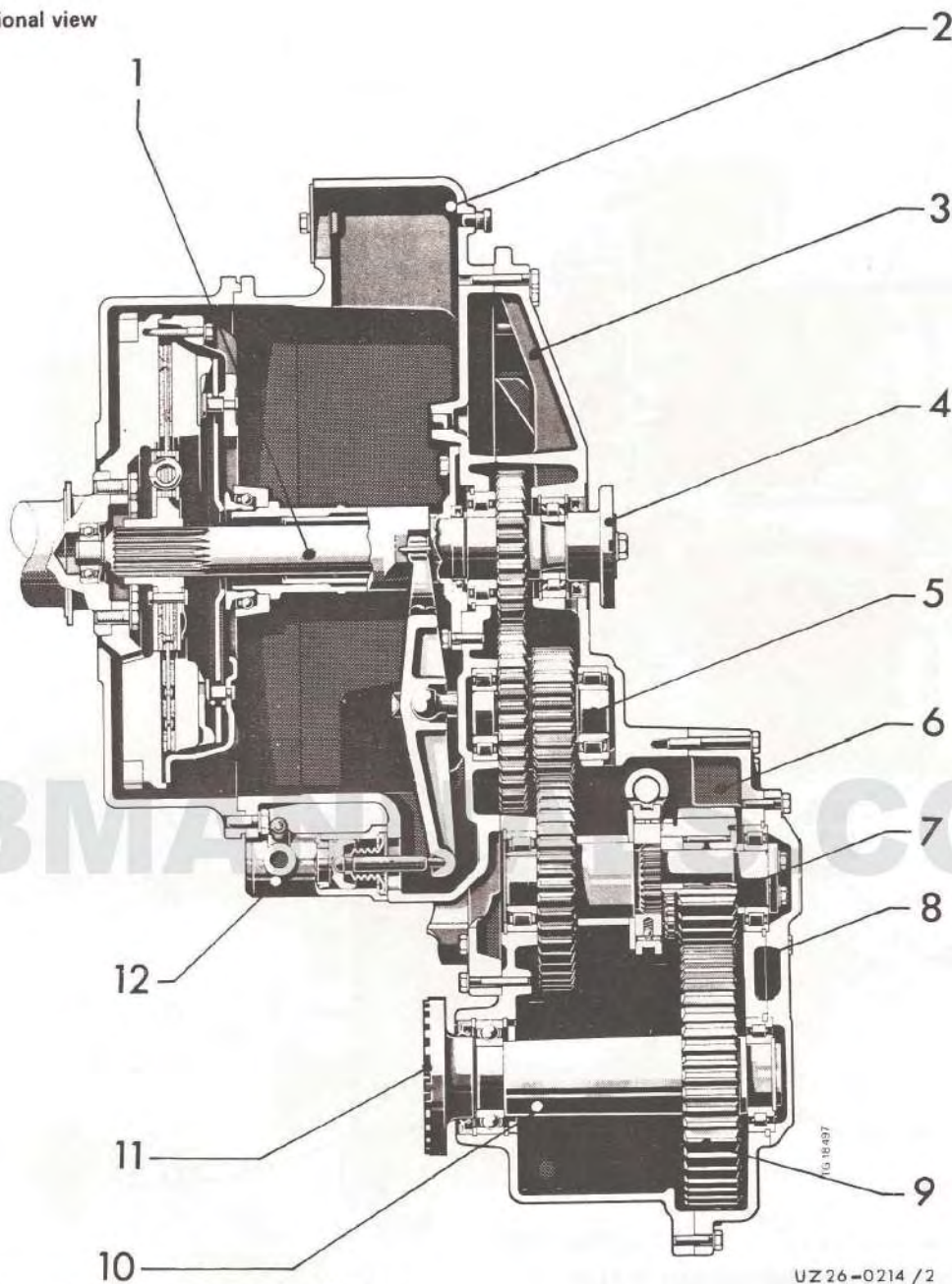
PTO transmission

Speeds, ratios

Chassis Model	Sales designation	PTO shaft		Ratios i
		Engine speed 1/min	PTO shaft speed 1/min	
435.111	U 1700 L	2 330	540	4.32
.113	U 1700 L/38	2 800	650	
.115	U 1300 L	2 200	540	4.08 ¹⁾
.117	U 1300 L/37	2 800	690	

1) approx. from chassis end No. 072 513

Sectional view



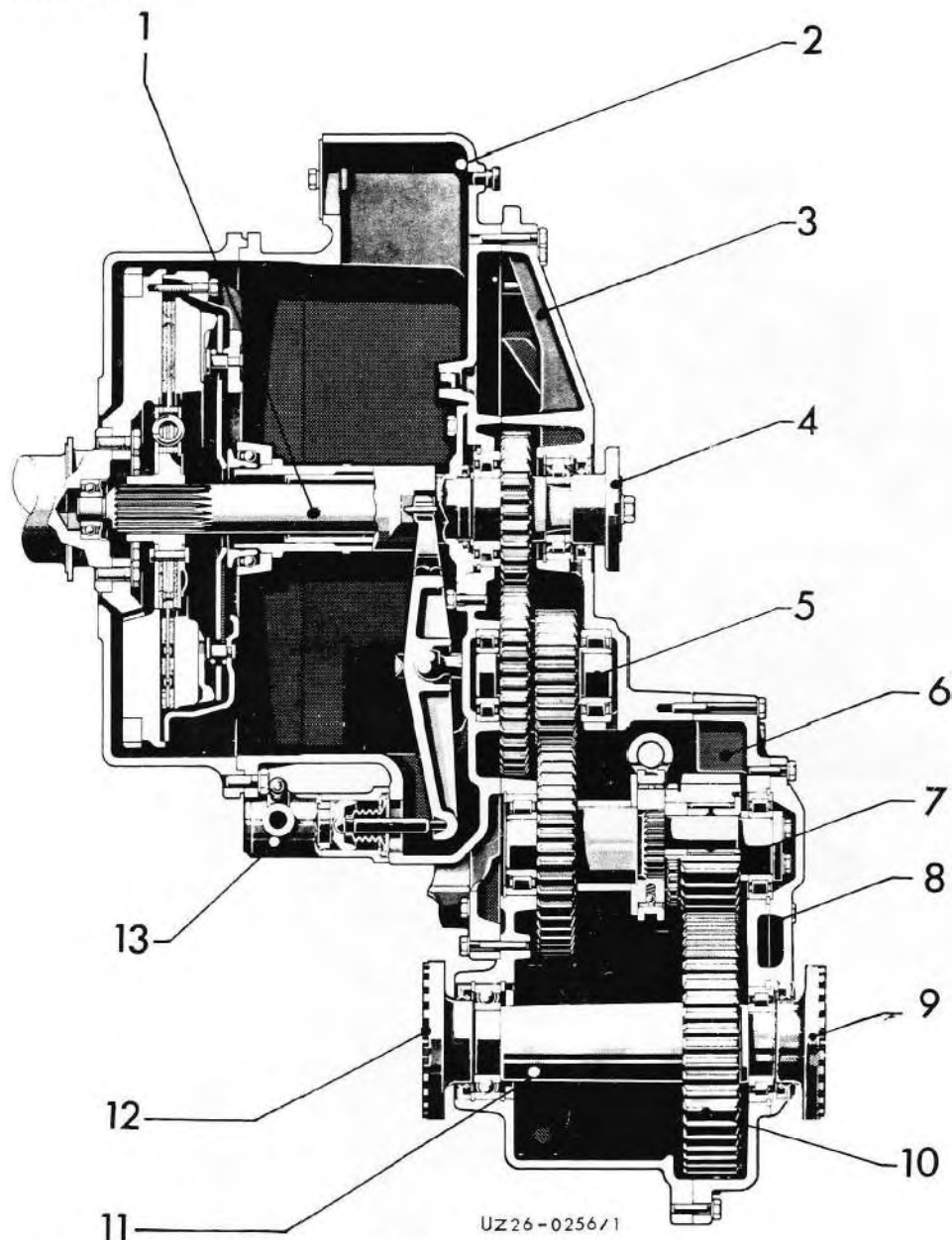
PTO transmission shaft, power take-off front

- 1 Input shaft
- 2 Clutch casing
- 3 Transmission housing
- 4 Main transmission output
- 5 Idler gears
- 6 Housing cover

- 7 Input shaft
- 8 End cover
- 9 Output gear 540/min
- 10 Output shaft
- 11 Power take-off front
- 12 Clutch slave cylinder

UZ 26-0214 / 2

Sectional view



PTO transmission shaft, power take-off front and rear

- | | |
|----------------------------|--------------------------|
| 1 Input shaft | 8 End cover |
| 2 Clutch casing | 9 Power take-off rear |
| 3 Transmission housing | 10 Output gear 540/min |
| 4 Main transmission output | 11 Output shaft |
| 5 Idler gears | 12 Power take-off front |
| 6 Housing cover | 13 Clutch slave cylinder |
| 7 Input shaft | |

Capacities

Assembly	Service product	SAE class	Quantity l
PTO transmission	Gear oil	80	5.75
	Engine oil ¹⁾	10 W	

¹⁾ For non-standard models

Special tools

Cons.No.	Tool	Number	Tool set
1	Basic drift for special tools	406 589 04 15 00	B
2	Fitter for shaft seals at output end of main transmission input shaft, with tool No. 1	435 589 01 43 00	B
3	Fitter for shaft seals at output end of PTO front, with tool No. 1	435 589 02 43 00	B
4	Fitter for shaft seals at output end of PTO front and rear, without fording capability, with tool No. 1	425 589 09 43 00	B
5	Suspension lug to remove PTO transmission	435 589 01 62 00	B
6	Basic drift for special tools	385 589 00 15 00	C
7	Fitter for shaft seals in shift shaft bores, with tool No. 5	425 589 07 43 00	C
8	Fitter for shaft seal in guide flange, with tool No. 1	425 589 10 43 00	C

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Necessary materials

Cons.No.	Description	Part No.
1	Teroson Atmosit, non-curing	regular
2	Longlife lubricating grease	regular
3	Molybdenum disulfide grinding paste	regular
4	Loctite 573	001 989 45 20
5	Loctite 241 ¹⁾	002 989 70 71

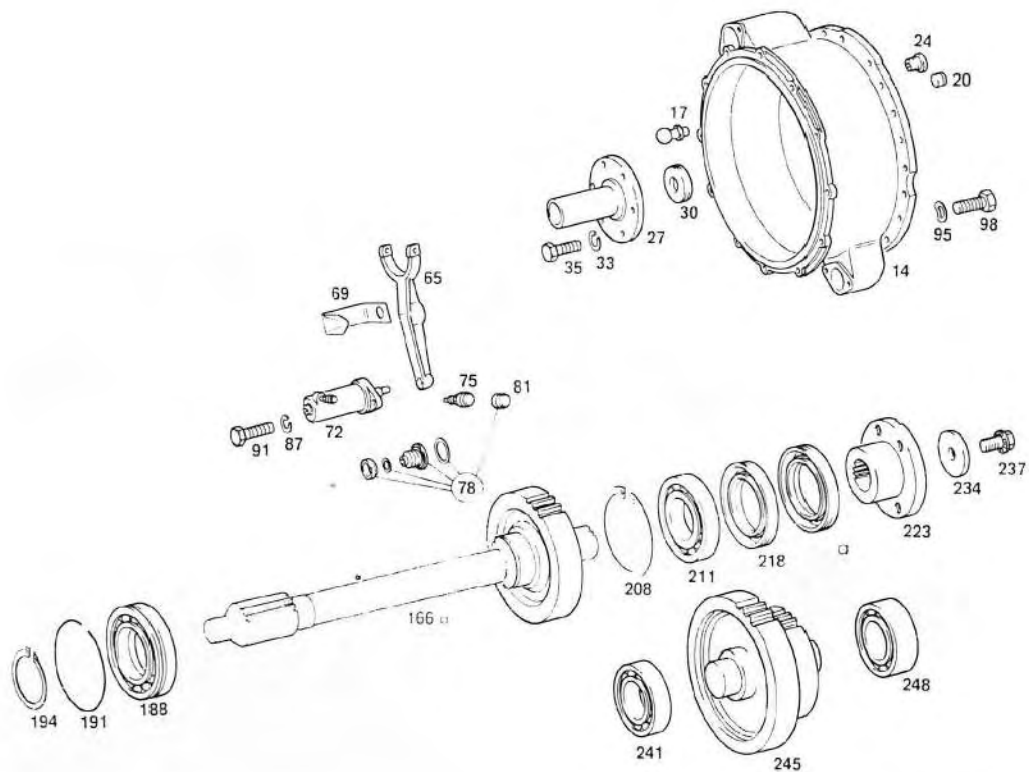
¹⁾ To seal all through-bolts

Tightening torques

Designation	Thread	Nm
Mounting, PTO output flange	M 30 x 1,5	300
Body support for rear cab mounting	M 20 x 1,5 - 8,8	410
Engine mounts at rear left and right on frame	M 14 x 1,5 - 10,9	140 ± 20
Mounting, main transmission output flange	M 10 x 30 - 12,9	92
PTO transmission to flywheel housing	M 10 x 30 - 8,8	42
Supporting disk to input shaft	M 8 x 30 - 12,9	45
Transmission housing to clutch casing	M 8 x 45 - 8,8	21
Housing cover to transmission housing	M 8 x 65 - 8,8	21
	M 8 x 30 - 8,8	21
End cover to housing cover	M 8 x 30 - 8,8	21
Guide flange in clutch casing	M 8 x 25 - 8,8	21
Slave cylinder to clutch casing	M 8 x 25 - 8,8	21

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Exploded view

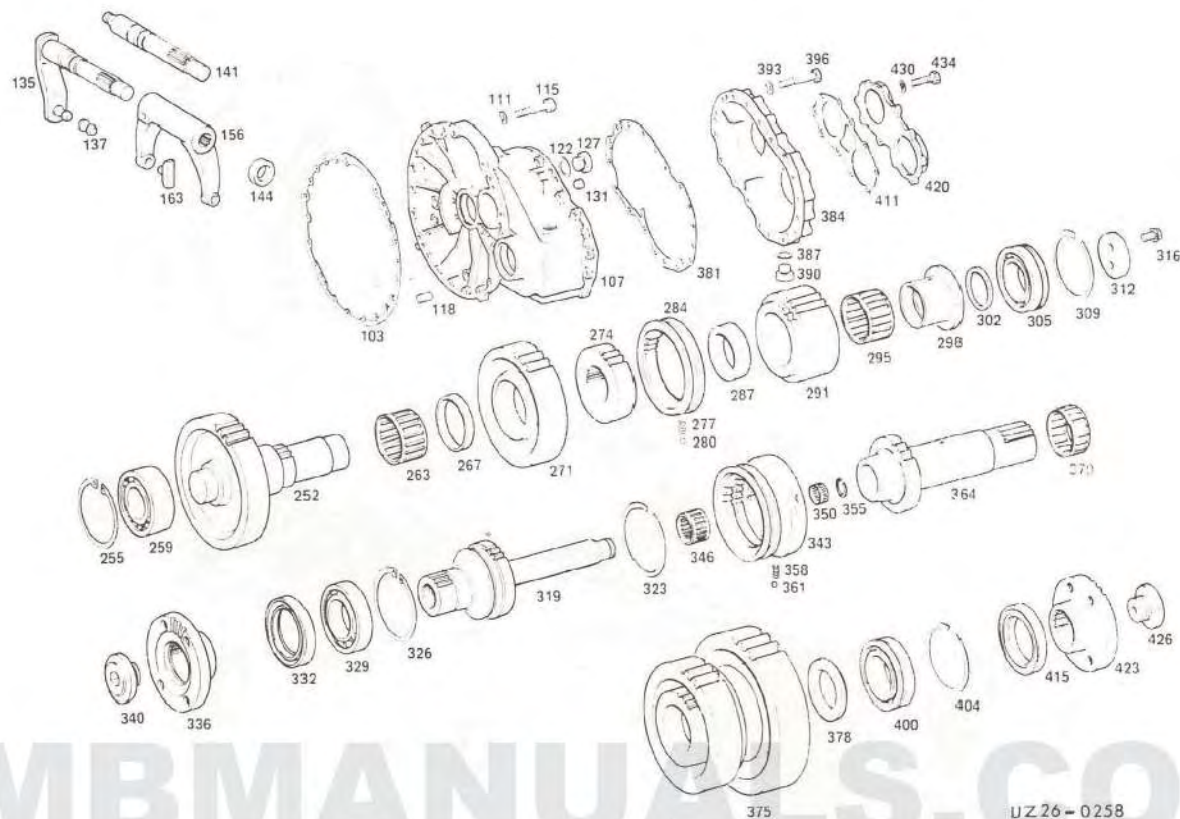


UZ 26-0259/1

PTO transmission shaft

- | | |
|-----------------------------------|--------------------------------|
| 14 Clutch casing | 166 Input shaft |
| 17 Knuckle pin | 188 Needle bearing |
| 20 Stopper | 191 Lock washer |
| 24 Breather | 194 Snap ring |
| 27 Guide | 208 Snap ring |
| 30 Sealing ring | 211 Deep-groove ball bearing |
| 33 Lock washer | 218 Sealing ring |
| 35 Bolt | 223 Flange |
| 65 Throwout lever for main clutch | 234 Washer |
| 69 Bracket | 237 Bolt |
| 72 Slave cylinder | 241 Cylindrical roller bearing |
| 75 Valve | 245 Intermediate gears |
| 78 Repair set for slave cylinder | 248 Cylindrical roller bearing |
| 81 Dust cap | |
| 87 Lock washer | |
| 91 Bolt | |
| 95 Spring washer | |
| 98 Bolt | |

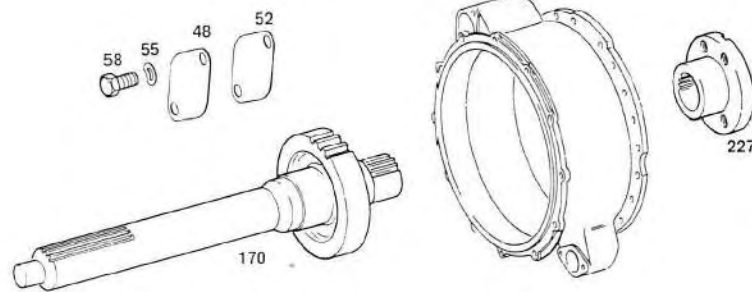
Exploded view



PTO transmission shaft

103	Gasket	280	Ball	358	Spring
107	Transmission housing	284	Sliding shift collar	361	Ball
111	Spring washer	287	Spacing ring	364	Hollow output shaft
115	Bolt	291	Drive gear 540/min	370	Needle bearing
118	Straight pin	295	Needle bearing	375	Output gear 540/1000/min
122	Sealing ring	298	Bushing	378	Spacing ring
127	Screw plug	302	Spacing ring	381	Gasket
131	Stopper	305	Cylindrical roller bearing	384	Housing cover
135	Shift shaft	309	Lock washer	387	Sealing ring
137	Knuckle pin	312	Washer	390	Screw plug
141	Shift shaft	316	Bolt	393	Spring washer
144	Sealing ring	319	Output shaft	396	Bolt
156	Dog	323	Snap ring	400	Cylindrical roller bearing
163	Sliding block	326	Snap ring	404	Lock washer
252	Input shaft	329	Deep-groove ball bearing	411	Gasket
255	Snap ring	332	Sealing ring	415	Sealing ring
259	Cylindrical roller bearing	336	Flange	420	End cover
263	Needle bearing	340	Screw plug	423	Flange
267	Spacing ring	343	Sliding shift collar	426	Screw plug
271	Drive gear 1000/min	346	Needle bearing	430	Spring washer
274	Dog ring	350	Needle bearing	434	Bolt
277	Spring	355	Lock washer		

Exploded view



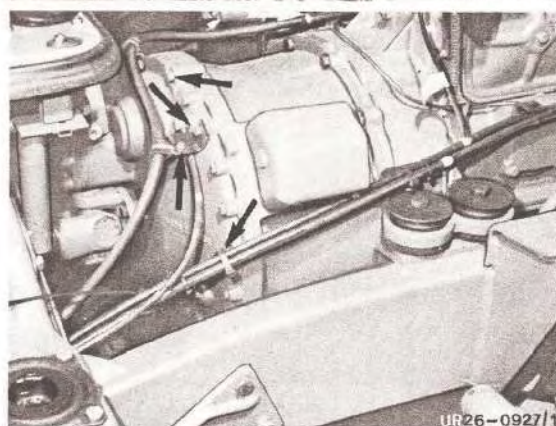
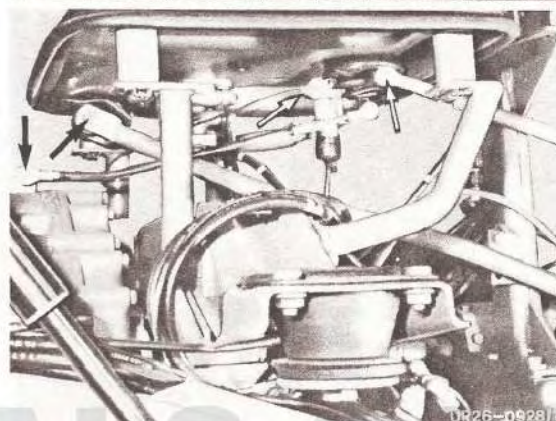
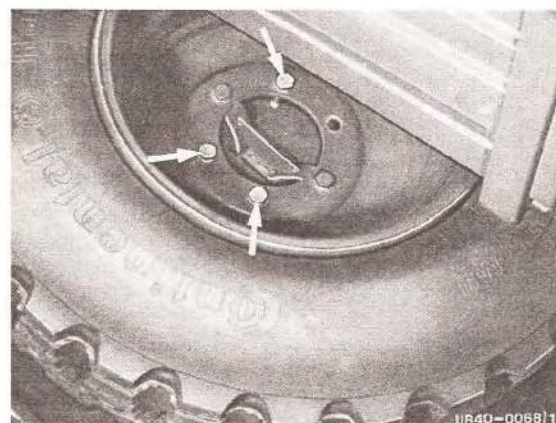
UZ26-0257

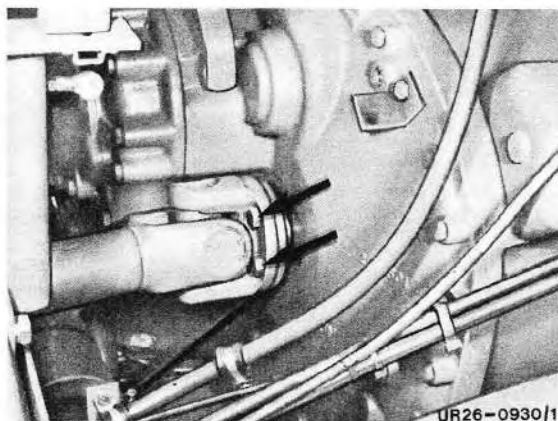
PTO transmission shaft

- 48 Metal cover
- 52 Gasket
- 55 Spring washer
- 58 Bolt
- 170 Input shaft
- 227 Flange

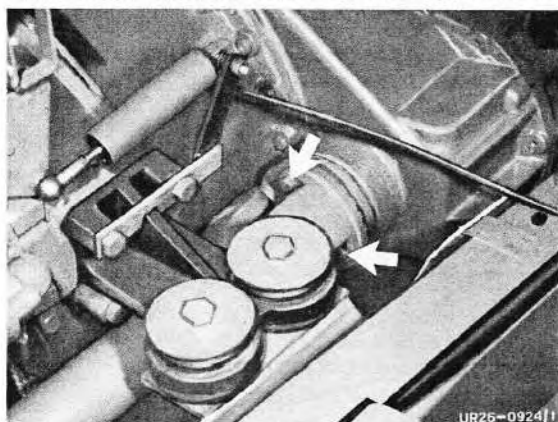
Removal

- 1 Raise cab,
referring to 60.5-2.1/1, or 60.5-2.2/1.
- 2 Remove spare wheel.
- 3 Detach main shift linkage, forward/reverse and
PTO shift linkage. Detach starting pilot line.
- 4 Remove body support mounting and slide to
rear.
- 5 Detach main battery cable and PTO transmis-
sion breather.





- 6 Remove transmission input shaft.



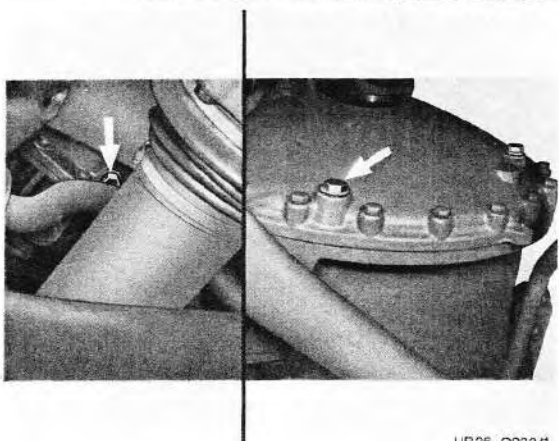
- 7 Detach PTO shaft at front.

Note: If there is a PTO shaft at both front and rear, remove both.



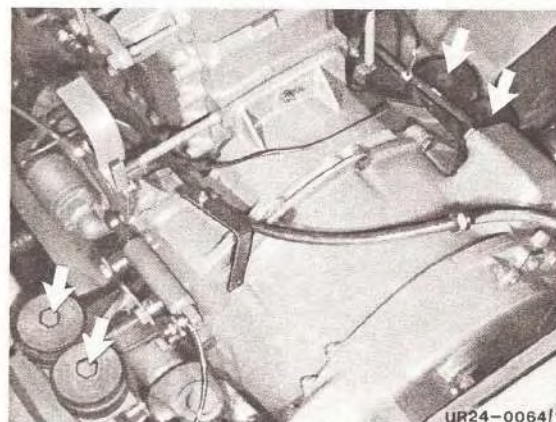
- 8 Detach clutch slave cylinder.

Note: Do not unscrew hydraulic lines.

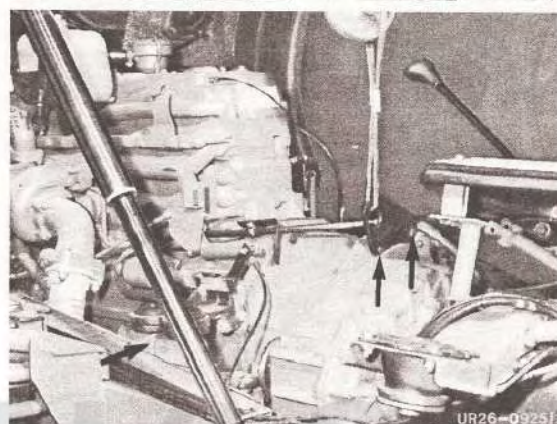


- 9 Drain oil.

10 Unscrew engine mount fastening bolts at left and right.



11 Attach special tool No. 5, hook PTO transmission with lifting tackle onto crane, lift slightly and insert wood blocks beneath engine mounts at left and right.



12 Unscrew PTO transmission, draw out to rear and then lower.

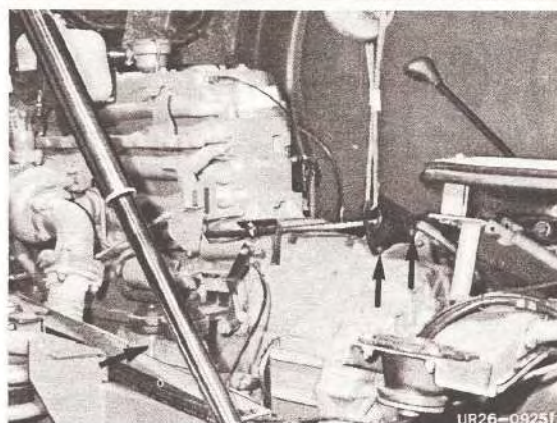


Installation

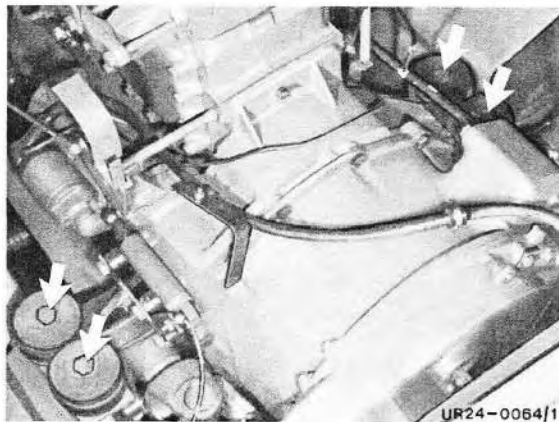
1 Coat PTO transmission with sealant No. 4, lift up and bolt to flywheel housing. See 1,4/1 for tightening torques.

Note: Coat bolts with locking fluid No. 5.

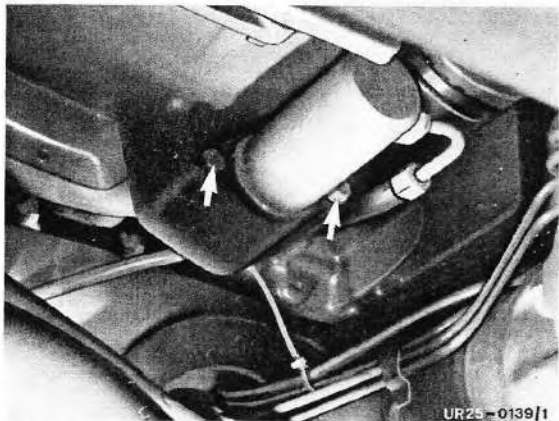
2 Lift up PTO transmission, remove wood block at left and right, and unscrew special tool No. 5.



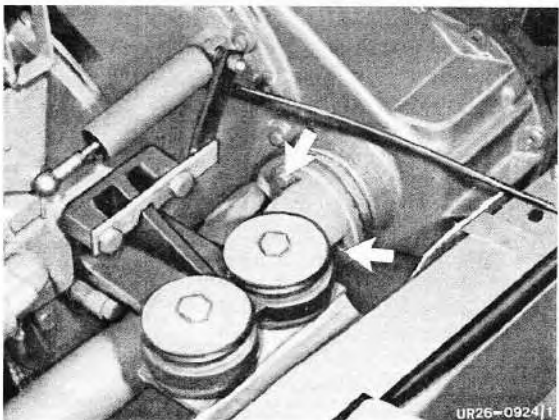
SA 35 737



- 3 Insert and tighten engine mount fastening bolts at left and right.
See 1.4/1 for tightening torques.

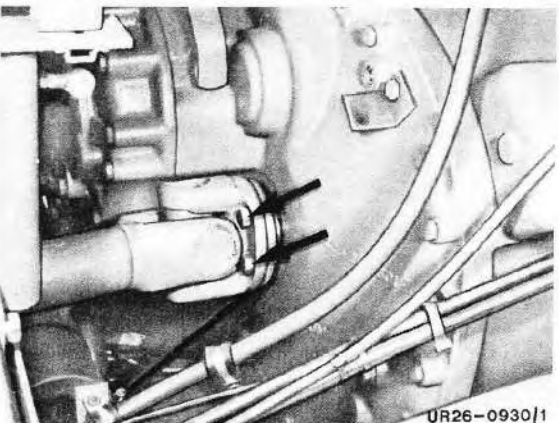


- 4 Coat clutch slave cylinder with sealant No. 4, inserting and tightening bolts with locking fluid No.5.
See 1.4/1 for tightening torques.



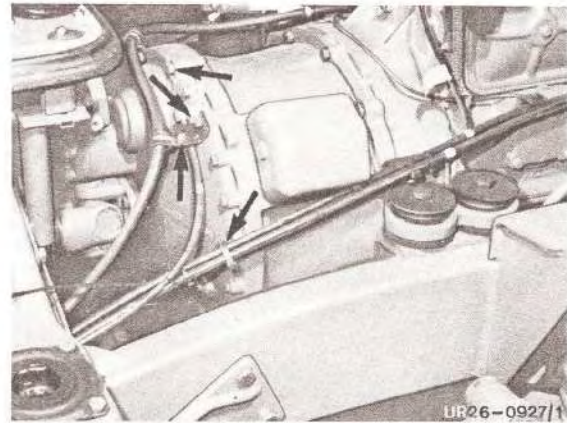
- 5 Attach and tighten PTO shaft at front, see 1.4/1 for tightening torques.

Note: If there is a PTO shaft at both front and rear, attach both.



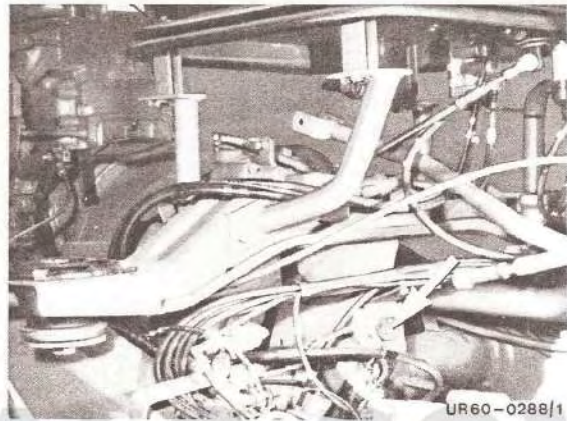
- 6 Attach and tighten transmission input shaft, referring to 1.4/1 for tightening torque.

- 7 Attach PTO transmission breather and main battery cable.

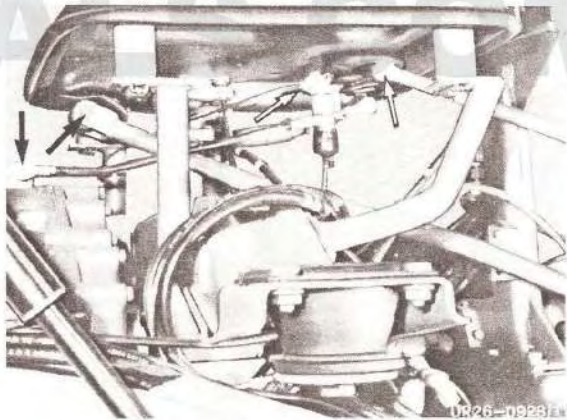


- 8 Attach and tighten body support, referring to 1.4/1 for tightening torques.

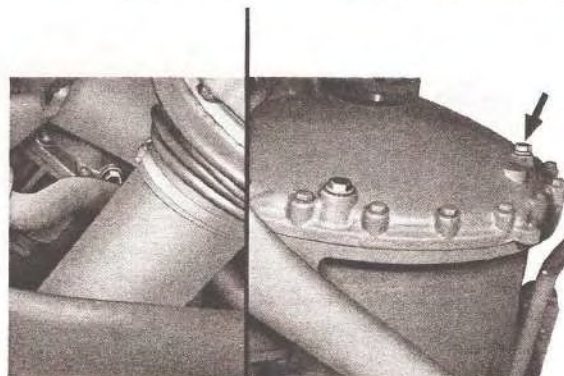
Note: Self-locking bolts are to be used once only.



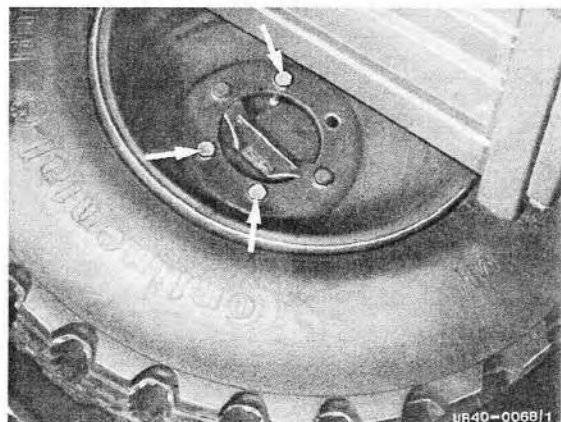
- 9 Attach main shift linkage, forward/reverse speed and PTO shaft linkage, fasten starting pilot line.



- 10 Fill transmission with oil, referring to 1.3/1 for quantity.



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11 Attach spare wheel.

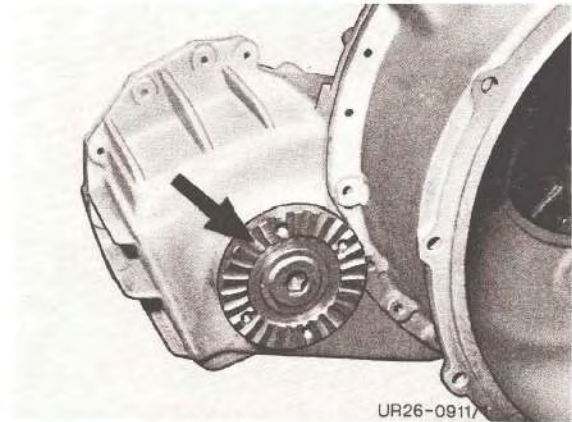
12 Lower cab,
referring to 60.5-2.1/1 or 60.5-2.2/1.

Disassembly

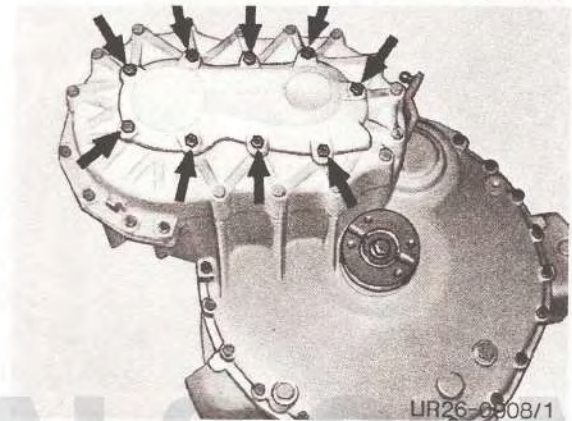
1 Referring to 2.1/1, remove PTO transmission and clamp in trestle.

Valid for power take-off at front.

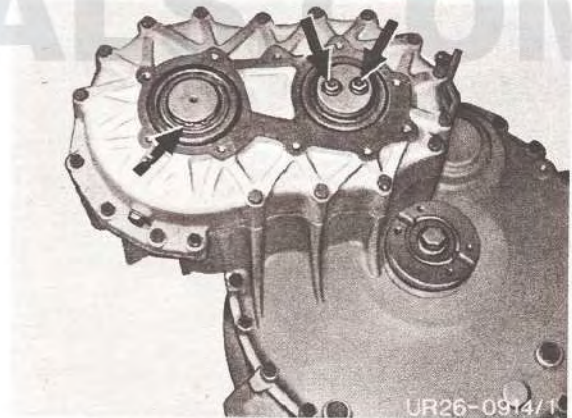
- 2 Detach flange for front PTO shaft.
3 Turn PTO transmission by 90°.



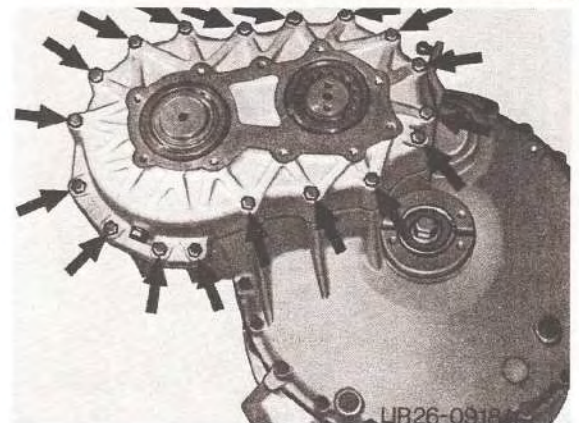
- 4 Detach end cover.



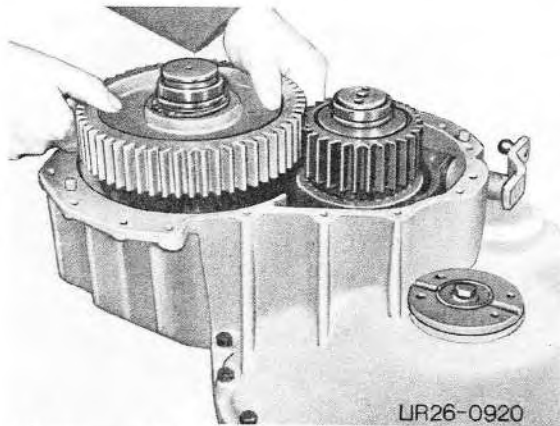
- 5 Remove supporting disk and snap ring.



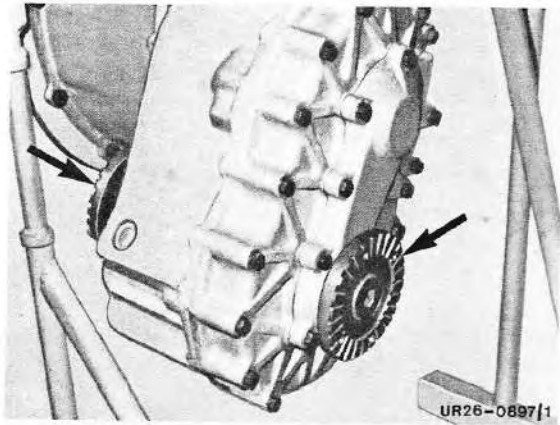
- 6 Remove housing cover and roller bearing.



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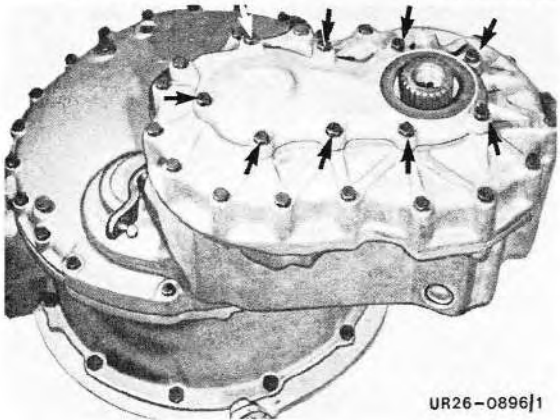


- 7 Remove output shaft.

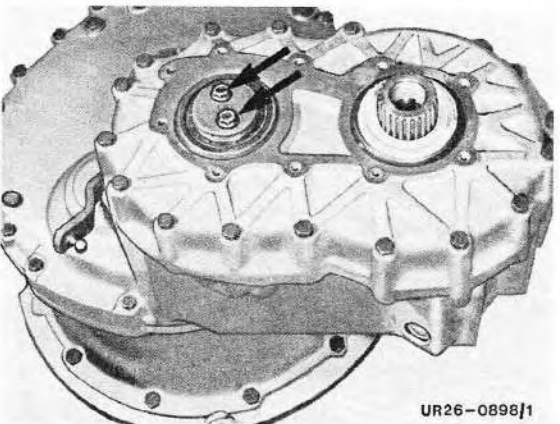


Valid for power take-off at front and rear

- 8 Detach flange for front and rear PTO shafts.
9 Turn PTO transmission by 90°.

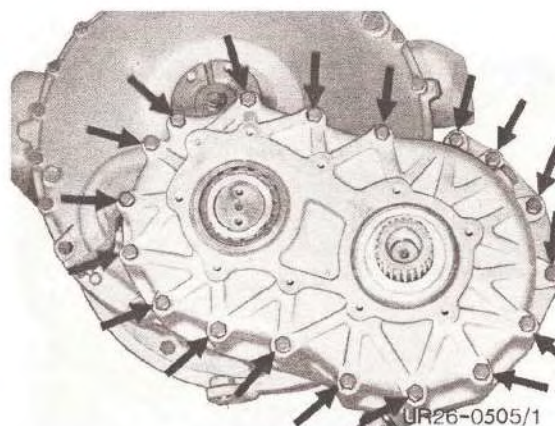


- 10 Detach end cover and remove shaft seal.

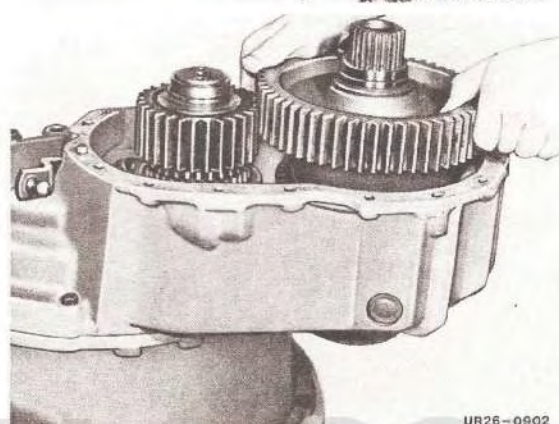


- 11 Remove supporting disk.

- 12 Remove housing cover and roller bearing.



- 13 Remove output shaft.

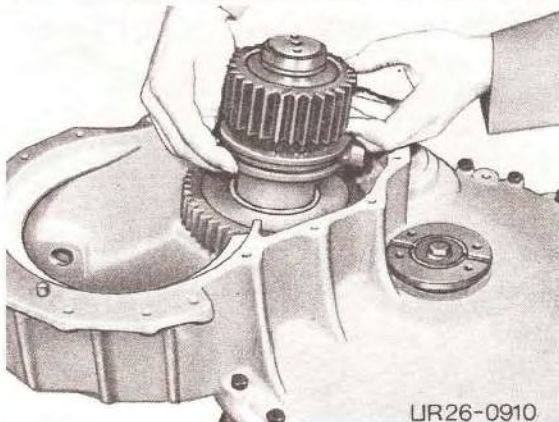


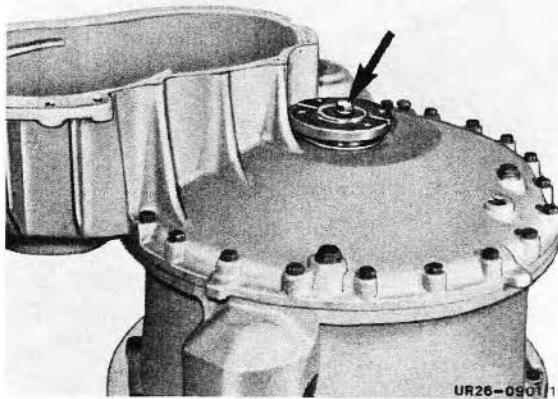
Valid for both versions

- 14 Release and remove shift shaft.

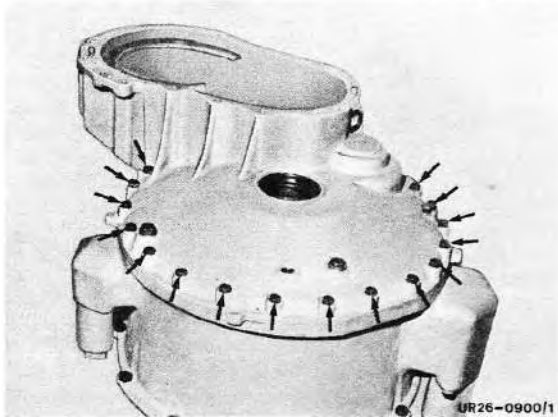


- 15 Remove input shaft.

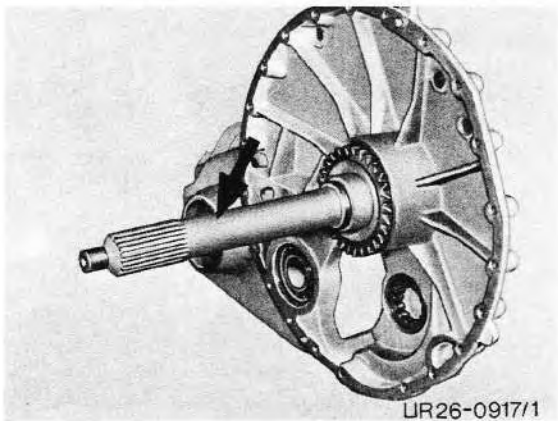




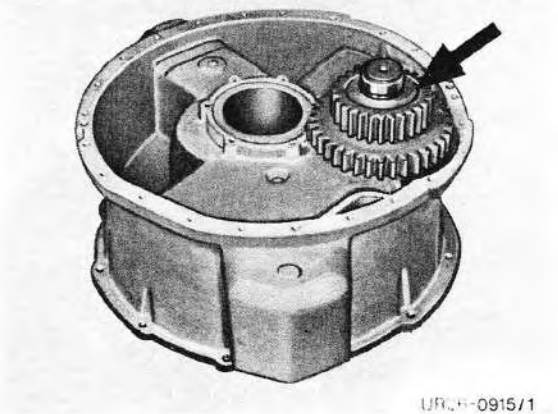
- 16 Remove supporting disk.
Detach output flange and shaft seal.



- 17 Remove transmission housing, as well as shaft seals and roller bearing if necessary.

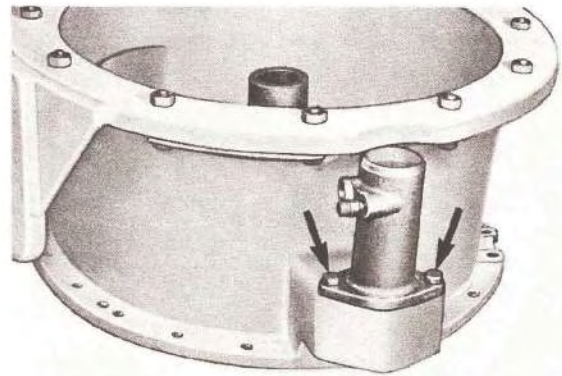


- 18 Force out transmission input shaft.



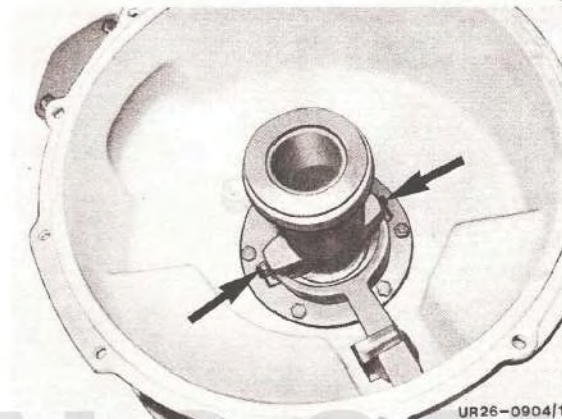
- 19 Remove idler gears.
20 Turn clutch casing by 180°.

- 21 If necessary, detach clutch slave cylinder.



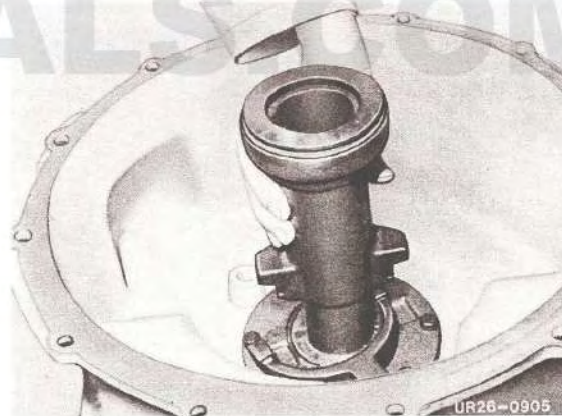
UR26-0903/1

- 22 Remove spring from throwout lever.



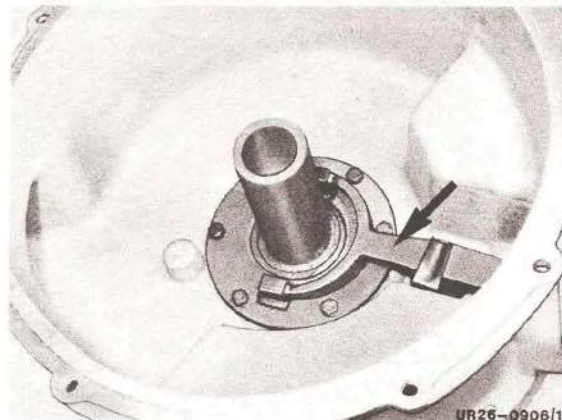
UR26-0904/1

- 23 Remove throwout bearing.



UR26-0905

- 24 Remove main clutch throwout lever.



UR26-0906/1